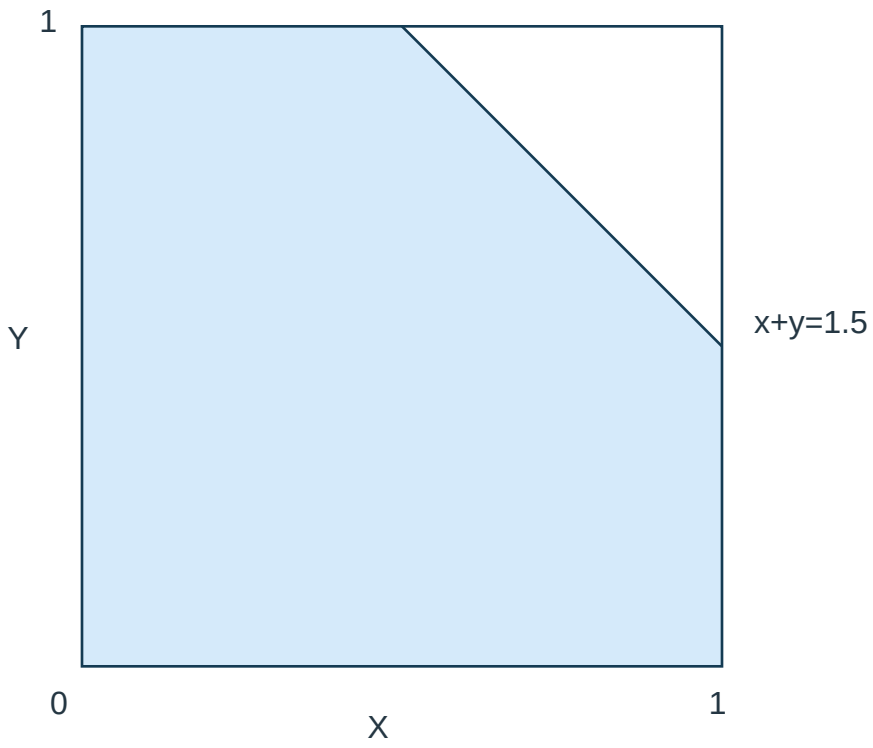
Sum of two Uniform(0,1) variables

Independent inputs X and Y have joint density 1 on the unit square.
For fixed u, X+Y <= u selects a region of possible input pairs.
At u=1.5 the selected region is a pentagon, not a quadrilateral.



F(u)=0 for u<=0; u^2/2 for 0<u<=1;
F(u)=1-(2-u)^2/2 for 1<u<2; F(u)=1 for u>=2.
With nonuniform joint density, integrate density over the region.
Area alone is probability only because the density here equals 1.

A bound that the text layer cannot verify

For Y uniform on [-1,1] and 0<u<1:

P(Y^2 <= u) = P(-sqrt(u) <= Y <= sqrt(u)).

The lower bound below is drawn as strokes; it is absent from extracted text.

$$\text{integral}_{-\sqrt{u}}^{\sqrt{u}} (1/2)\, dy$$

The correct result is sqrt(u). At u=1/4 it is 1/2.

The input event is one-dimensional; the graph of y^2 is a different object.

If the page cannot be rendered, do not infer that the author omitted a sign.

Two explicit Normal conventions

Source A defines Normal(mu, sigma): the second parameter is SD.

A: mean 5, SD 2 is Normal(5,2), with variance 4.

Source B defines N(mu, v): the second parameter is variance.

B: mean 5, SD 2 is N(5,4). These describe the same distribution.

Never transfer a second parameter without checking the source convention.

Chapter 7: proof-first outline

1. Moment-generating functions and Taylor expansion near zero.
2. Derivation for standardized sums under the stated MGF assumptions.
3. The iid finite-variance CLT statement; applications and limitations.
4. Simulating R independent samples of n observations each.

Increasing n changes the distribution of a sample mean.

Increasing R refines simulation of that fixed distribution.

The CLT alone gives no monotonic finite-n approximation-error guarantee.

Under a short application goal, defer the proof explicitly; its prerequisites remain relevant to the proof-first lecture.

Zero pairwise covariances suffice for variance additivity.

Independence is sufficient too; it is not necessary.

Example: X uniform on {-1,0,1}, Y=X^2. They are dependent.

E[X]=0, E[Y]=2/3 and E[XY]=E[X^3]=0, hence Cov(X,Y)=0.

Var(X)=2/3; Var(Y)=2/9; Var(X+Y)=8/9.